Final Project

Please analyze the assigned mortgage-back security and answer the questions below. This is an individual project, DO NOT WORK IN GROUPS. Totals: 100 points

Please follow the lecture slides carefully, show every step in your calculations, and present each cash flow in accordance with the lecture slides so that your analysis is fully transparent. Avoid prebuilt black-box functions to enhance the learning experience. Use a discrete, rather than a continuous, framework, as we focus on empirical implementations.

Part I. Pricing Agency-type MBS

Use the assigned security. Assume that full and timely payment of scheduled principal and interest on this security is guaranteed by the U.S. government.

1. Generate a large number of paths of interest rates at the frequency and to horizon desired. You may refer to Fabozzi, p. 383 regarding approaches to generating interest rate paths.

2. Suggest and justify a prepayment model, e.g. a PSA model appropriate for your security.

3. Calculate the cash flows of the security along each path.

4. Compute the value of the security as the average of the discounted values across paths. Comment on your finding. Is your security fairly priced? Are you sure?

Part II. Pricing Private-label-type MBS

Use the assigned security. Now assume that the timely payment of scheduled principal and interest on this security is guaranteed by a private financial institution that has been assigned a credit rating of **A** by S&P Global. Compute the value of the security using the approach specified in Steps 1–4 above, but now also adjust for credit risk. Use the expected cash flow approach covered in Topic 5, which relies on probabilities of default, probability of survival, and recovery rates. Comment on your findings.

Important: *Your submission should include the cover page, a summary of all your findings and supporting materials such as data files, screenshots of the original valid data sources , codes in PDF format and/or Excel spreadsheets*